PROJECT MILESTONE 02

Real-Time Smart Room Monitoring & Data Pipeline

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Data Stream Collection & Simulation

Primary Dataset

UCI Occupancy Detection: 20,560 sensor readings from a single office room.

Recorded Features

Temp, Humidity, Light, CO_2, and Humidity Ratio recorded every ~1 minute.

Data spans Feb 4–18, 2015, split into training (8,143 records) and two test sets.

Stream Simulation: Replaying records one-by-one in timestamp order.

SMART ROOM MONITORING ENVIRONMENT

Pipeline Implementation Architecture

Core Environment

Built in Python using the standard library. Zero external dependencies for maximum portability.

Data Ingestion

Simulates real-time sensor arrival by reading and parsing records one at a time.

Sliding Window Logic

Maintains 30 records (~30m) using deque. True sliding window with auto-eviction.

Real-time Processing: Recomputing mean, std, min, and max for every window update.

Preliminary Results & Observations

Training Metrics

8,143 records processed. Room was occupied only 21.2% of the time, creating a skewed distribution.

Strongest Indicators

Light signal is binary: 0 lux (unoccupied) vs 400-600+ lux (occupied). CO_2 shows clear, rising trends.

Subtle Variables

Temperature and Humidity exhibit slower, subtler shifts compared to the sharp spikes of light and gas.

Real-Time Detection: Rolling window statistics successfully capture transitions as the stream progresses.